ABHINAV ANAND

Software Engineer Intern (Python / Backend) | Applied AI & ML

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SUMMARY

B.Tech CSE (AI/ML) student building Python backend and applied AI systems with FastAPI, Docker, and end-to-end ML, NLP, and computer-vision pipelines. Built DocuShield, a document-forensics platform with a 90-test automated backend suite, plus two deployed applications, including an AI resume analyzer using Supabase and Groq (Llama 3).

TECHNICAL PROJECTS

DocuShield — Fake ID & Document Forensic Screening System

Smart India Hackathon 2026, PS SIH26188

Stack: Python, FastAPI, React (Vite), OpenCV, EasyOCR, Tesseract, Docker

- Built a multi-stage OCR workflow for 5 identity-document types (passport, visa, national ID, driving licence, permit), combining EasyOCR/Tesseract with ICAO MRZ parsing and checksum validation (including Verhoeff for Aadhaar); served it through FastAPI with a decoupled React (Vite) client.
- Analyzed potential image tampering using 3 forensic signals—Error Level Analysis, ORB + RANSAC copy-move detection, and EXIF metadata—in a rule-based evidence-fusion engine that produces an explainable authenticity risk score.
- Added a 90-test automated backend suite covering document verification and tamper-detection workflows.

AI Resume ATS Analyzer

Sep 2026

Stack: Python, FastAPI, Streamlit, spaCy, Sentence Transformers, Groq API (Llama 3), Supabase, Docker

- Built a 100-point resume-scoring engine covering formatting, keyword/skill matching, content quality, and ATS parseability, with a FastAPI backend and Streamlit frontend packaged in Docker.
- Built a skill-context validator that compares listed skills with project evidence using spaCy and Sentence Transformer embeddings; added automated personally identifiable information (PII) screening.
- Integrated the Groq API (Llama 3) to generate resume-improvement suggestions and used Supabase for authentication and persistent analysis history backed by PostgreSQL.

CreditWise — AI Loan Risk Intelligence System

Mar 2026

Stack: Python, Scikit-Learn, Pandas, NumPy, Streamlit

- Classified loan applicants into 3 risk tiers (Low/Medium/High), reporting 87.5% accuracy with a Scikit-learn multiclass classifier and Pandas-based feature engineering.
- Identified and removed a data-leakage issue from the feature set before release, then deployed the model as a Streamlit app for interactive what-if risk scoring.

TECHNICAL SKILLS

Languages:	Python, C++, JavaScript, SQL
Backend & Databases:	FastAPI, REST APIs, PostgreSQL, Supabase (Auth, Postgres)
Frontend & Apps:	React (Vite), Streamlit
Machine Learning & NLP:	Scikit-learn, Pandas, NumPy, spaCy, Sentence Transformers, Groq/Llama 3 APIs, Feature Engineering
Computer Vision & OCR:	OpenCV, EasyOCR, Tesseract, ORB + RANSAC, Error Level Analysis (ELA), ICAO MRZ Parsing
Tools & DevOps:	Docker, Git, GitHub, Linux Shell

CERTIFICATIONS & HACKATHONS

- **Smart India Hackathon 2026:** Qualified and built DocuShield for Problem Statement SIH26188.
- **HackerRank:** SQL (Advanced) Certified.
- **Infosys Springboard:** Python Certified.

EDUCATION

Lovely Professional University, Punjab, India Aug 2025 – Aug 2029

B.Tech in Computer Science and Engineering (AI & Machine Learning Specialization)